Self-Organized Criticality: Sandpile Model on Undirected Graph, Influence of Topology and Dissipation Rules on Avalanches

1. Introduction

Almost everything in nature tends to cluster around a center value. An example might be the height of humans. This is described by a normal distribution which itself got its name because this pattern is so often observed not only in nature but in human-made systems as well.

Contrary to this, there are systems which could not be described by normal distribution. One example are the forest fire sizes. If they were to adhere to normal distribution, there would be known an *average forest fire size*. Even though most of forest fires are small, there is nothing limiting the potential size. This is one of key properties of systems better described by power-law, scale-invariance.

Normal distribution is *heavy tailed*, meaning that most of values are distributed within small range around the center (mean) and values far from it are very impropable. The probability density function of normal distribution is described in Equation 1

$$\begin{aligned} f(x) &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \\ \mu &\ldots \text{ mean} \\ \sigma^2 &\ldots \text{ variance} \end{aligned} \tag{1}$$

TODO: plot normal dist?

As opposed to this, power-law[1] is *long tailed*. Its exponential character described in Equation 2 result in majority of the values being very far. That corresponds with the scale-invariance: Any (forest fire) size is realistically possible, even though smaller sizes are more likely. Power-law exhibit linear relationship between $\log f(x)$ and $\log x$. In a `log-log plot` it forms a straight line.

$$\begin{aligned} f(x) &= ax^{-k} \\ k &\ldots \text{ constant exponent} \end{aligned} \tag{2}$$

TODO: plot power-law? in normal and log-log plot

The emergence of power-law distribution is often connected to systems critical state. Consider phase transition of water from liquid to vapor. At the liquid-vapour boundary curve, both liquid and vapour can coexist but are distinctly separate. The boundary terminates at some critical temperature and critical pressure at which the critical point of a system is defined. While the system is in the critical state, the water is in not in any one state.

Systems in critical state are most responsive to input and behaves unpredictable. This is the case for phase transition and forest fires. Although, there is a significant difference. Criticality can be reached in many systems by fine-tuning certain parameters. In the case of phase transition the parameters were pressure and temperature. However, systems exist which reach criticality independently of parameters.

TODO NOTE?: usually high degree of freedom systems

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[1] Power-law on $(0, \infty)$ cannot be a probability distribution, because the areas near 0 and under the tail are infinite. However, with simple restrictions: $f(x) = ax^{-k}, \quad k > 1, \quad x > x_{\text{min}}$ it is a valid distribution.

The phenomenon of systems reaching criticality by themselves is called *self-organized criticality (SOC)*. The concept of SOC was first discovered in 1987 by Bak, Tang and Wiesenfeld and its properties were demonstrated on a sandpile model [1]. This model and its properties is the main focus of this thesis.

TODO: talk about the idea of putting it on random graphs, not just general?

The original sandpile model was a cellular automaton on a sqare and cubic lattice. However, the idea is to use another underlying graph structures for the automaton. Before this generalization we introduce few needed concepts from graph theory.

2. Graph Theory

An undirected graph $G = (V, E)$ is composed of set of elements (vertices) V and E is a set of unordered pairs $\{v_1, v_2\}, \quad v_i \in V$ called edges.

The *vertex degree* $d(v_i)$ is defined as the number of vertices connected to v_i via an edge.

Path length describes the distance between two vertices for a given path. The path length is equal to the number of vertices in between the first and last vertex on a path. The *shortest path length* refers to a distance between two vertices such as any other path between them produces at least the same path length. It is useful to also define an *average shortest path length*, an average of shortest paths between any two vertices in a graph.

TODO: small-world network properties (the previous needed for that)

3. Sandpile Model

TODO: where to put

This model was later generalized from square lattice to arbitrary graph [2]. That generalization is important because it allows us to study the model's behaviour for different graph topologies.

The original sandpile model presents a simple chessboard and falling sand grains. Each grain falls on a chess square determined by a random distribution. If the number of grains on any square reaches four, the sandpile topples and the four grains fall on adjacent squares, possibly causing additional squares to topple. In case the square does not have four adjacent squares (i.e. it is on the edge of chessboard) the remaining grains fall off the board and are therefore removed from the model.

The topology of this model can be expressed as an undirected graph, a square lattice of the size 8×8. However, it is not a correct representation until an additional vertex is introduced. The often called sink vertex has an important property that it cannot be toppled. When the sink is connected to every vertex on the edge of the square lattice (twice to corner vertices) the graph representation becomes valid[2]. This is a foundational observation, for it allows us to use any underlying graph for the sandpile model.

There are two possible state for any vertex. It is stable if it contains less grains than its degree. That is, if the number of edges from that vertex (including edges to sink) is higher than the number of grains. Otherwise the vertex is considered unstable. The system is stable if all vertices are stable.

If a grain is added to any stable vertex, it can destabilize it resulting in an avalanche. Avalanche is the set of topplings which occur before the system is stabilized again. One extreme case is an

[2]If a sink is connected to any vertex multiple times, the graph becomes multigraph. However, that is not a bad thing for this thesis where the important thing is that the topology itself is a graph. The different ways of connecting a sink to vertices are explored in Section 3.1. The word graph is used interchangibly with multipgraph.

avalanche of size one, which is only the origin of avalanche toppling. The other extreme for chessboard is an avalanche of size 64, which is the whole board toppled.

TODO: proof?

The process of a grain leaving the board is called dissipation and it is an important property of the model. Without dissipation the model supersaturates, leading to one infinite avalanche. For a connected graph (i.e. graph where for every pair of vertices exist at least one path from one to another) to dissipate at least one vertex connected to a sink is necessary. For any arbitrary graph the requirement heighten to every vertex being connected to a sink.

TODO: how can the model be extended, rules and graphs. foundation for it is in Section 3

3.1. Dissipation Rules

As discussed in Section 3 the existence of a sink vertex is necessary for any graph representation of a sandpile model. However there are various rules by which you can determine if a given vertex should be connected to a sink vertex and how many times if so.

For example the original model uses a rule (named by me as) *Fill to Four*. This rule states that the degree of vertex must be exactly four. The sink vertex is connected to all vertices as many times it is needed to fulfill that rule. For the square lattice, only vertices on the edge are connected (exactly once except for corner vertices, which are connected twice). This rule precisely reproduces the original chessboard idea.

For an arbitrary graph though, the rule is insufficient. It does not ensure dissipation which can lead to graph supersaturation and never-ending avalanche. This rule does not reflect on the vertex degree if it is higher than four.

One possible rule which guarantees dissipation on any graph may be *All Once*. The rule is fair in a sense that all vertices have connection to the sink and is uniform because every vertex, no matter its degree, is connected exactly once.

Finally, another rule which assures dissipation is *As Many As Neighbours*. This rule connects every vertex to sink proportionally to the number of its neighbouring vertices, therefore emphasizing how many connections it has.

Every rule represents a different approach and not all can be applied to all types of graph topology.

3.2. Underlying Graphs

Standard choice for graph in sandpile model is square lattice with arbitrary size.

TODO: square lattice - properties

TODO: ER graph - we don't have

TODO: WS graph - we have

4. Observed Behaviour

TODO: how did it work for different types of graphs and dissipation rules? also the graph configurations (eg β for WS)

TODO: metrics, how it changed criticality? origin distribution, avalanche size (max, avg)

5. Technical Documentation

TODO

Bibliography

[1] P. Bak, C. Tang, and K. Wiesenfeld, "Self-organized criticality: An explanation of the 1/f noise," *Physical Review Letters*, vol. 59, no. 4, pp. 381–384, July 1987, doi: 10.1103/PhysRevLett.59.381.

[2] A. E. Holroyd, L. Levine, K. Mészáros, Y. Peres, J. Propp, and D. B. Wilson, "Chip-Firing and Rotor-Routing on Directed Graphs," in *In and Out of Equilibrium 2*, Birkhäuser Basel, 2008, pp. 331–364. doi: 10.1007/978-3-7643-8786-0_17.

TODO

- multigraph?

TODO PS

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